

ASTROPHYSICS FOR PEOPLE IN A HURRY



A Cosmic Zoom from the Big Bang to the Present

Based on the work of Neil deGrasse Tyson

**The universe is under
no obligation to make
sense to you.**

– Neil deGrasse Tyson

We are about to cover 14 billion years of history. The concepts are vast, but the story is specific. Let's begin at $t=0$.

TIME: $t=0$ to $t=10^{-43}$ seconds

The Planck Era

In the beginning, all space, matter, and energy were contained in a volume less than one-trillionth the size of a period. Conditions were so hot that the four forces of nature were unified.

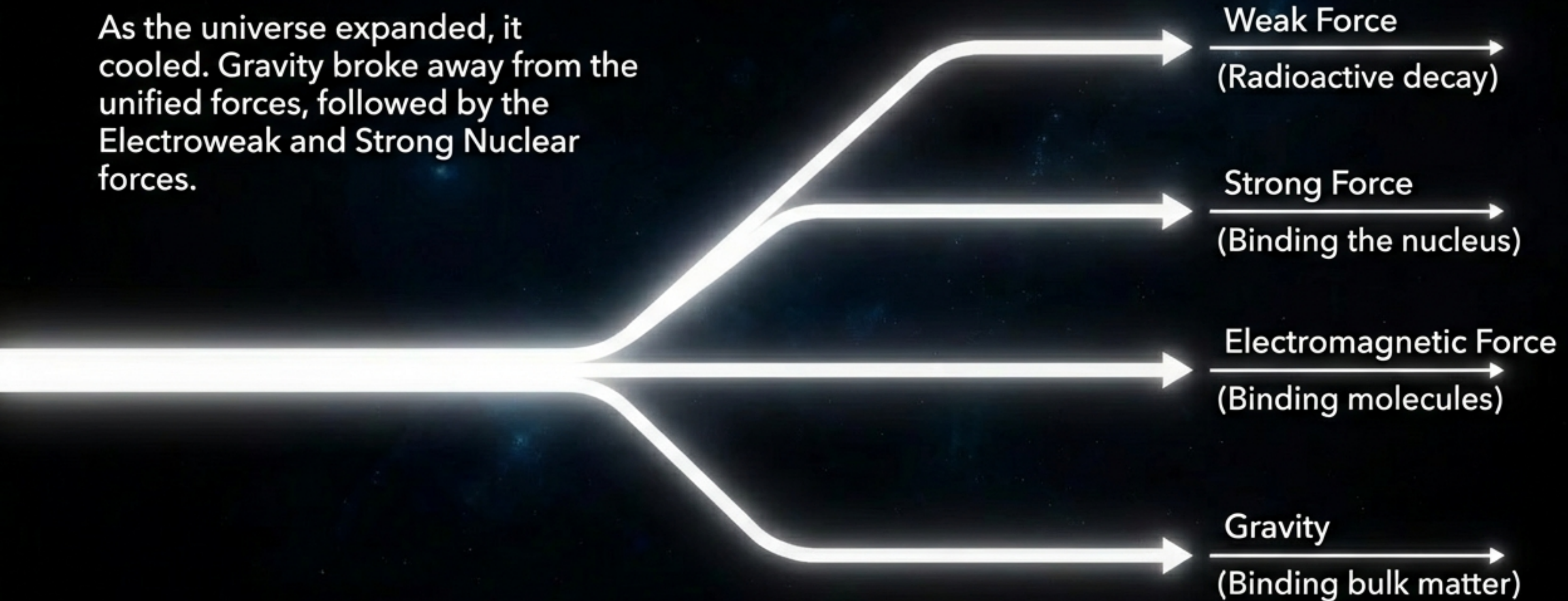


Gravity and Quantum Mechanics forced together. The large was small, and they were one.

TIME: $t=10^{-35}$ seconds

The Great Split

As the universe expanded, it cooled. Gravity broke away from the unified forces, followed by the Electroweak and Strong Nuclear forces.



By the end of this split, the four forces we know and love were distinct.

The One-in-a-Billion Survivor



Antimatter:
1,000,000,000

Matter:
1,000,000,001

The universe was a seething soup of quarks and leptons. Matter and Antimatter were created in pairs and immediately annihilated each other.

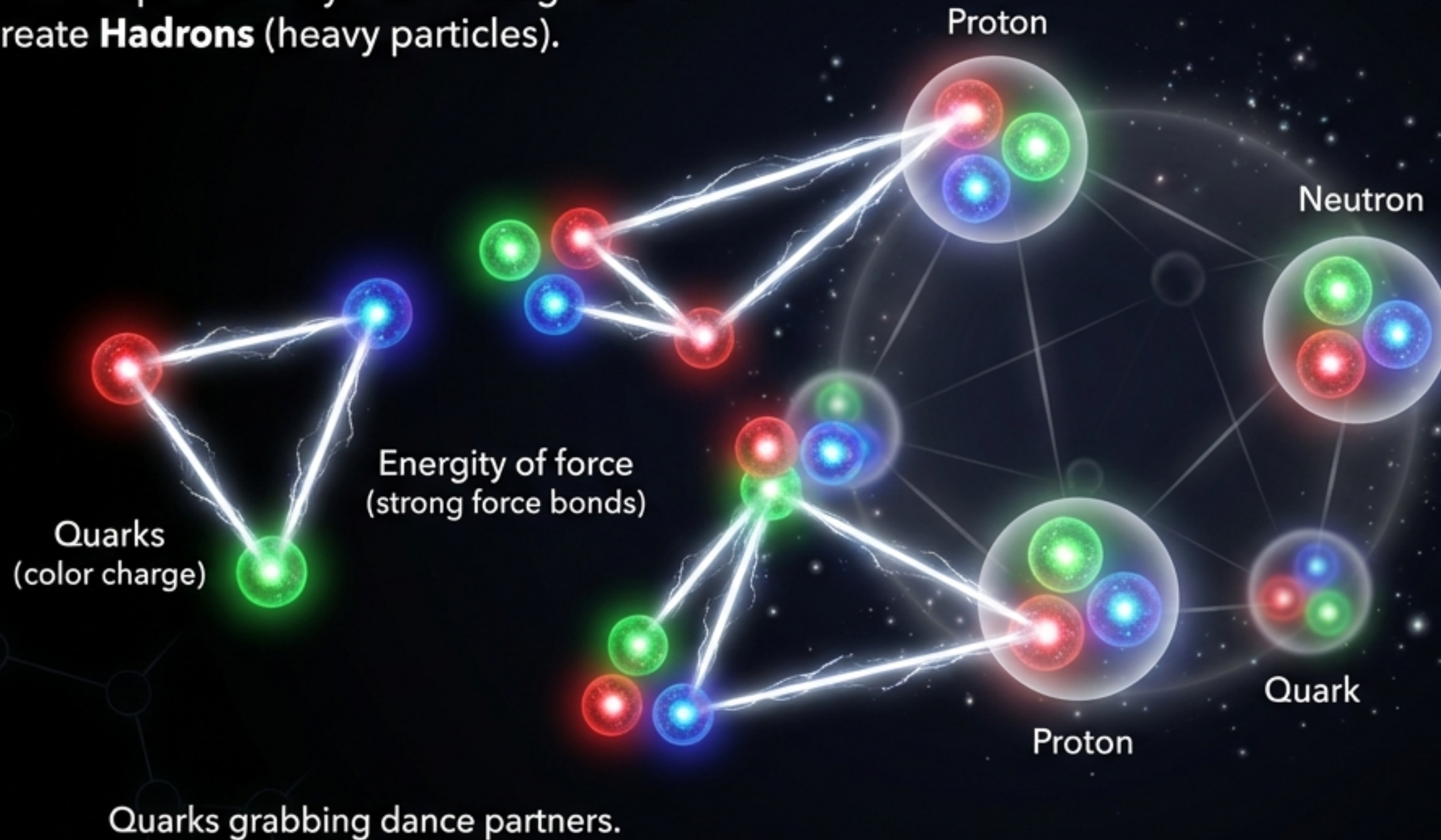
A slight asymmetry emerged.

That single “odd man out” is the source of all matter in the universe today—every galaxy, star, planet, and petunia.

The Hadron Era

The universe was no longer hot enough to cook quarks. They bonded together to create **Hadrons** (heavy particles).

TIME: $t=10^{-6}$ seconds



This transition birthed the protons and neutrons that make up atomic nuclei.

Physicists at CERN use the **Large Hadron Collider** to smash particles together just to recreate the conditions of this specific microsecond.



CERN / Large Hadron Collider

TIME: $t=2$ minutes

The Freeze

Protons fused with neutrons to form atomic nuclei.
Then, the universe cooled below the threshold for fusion.

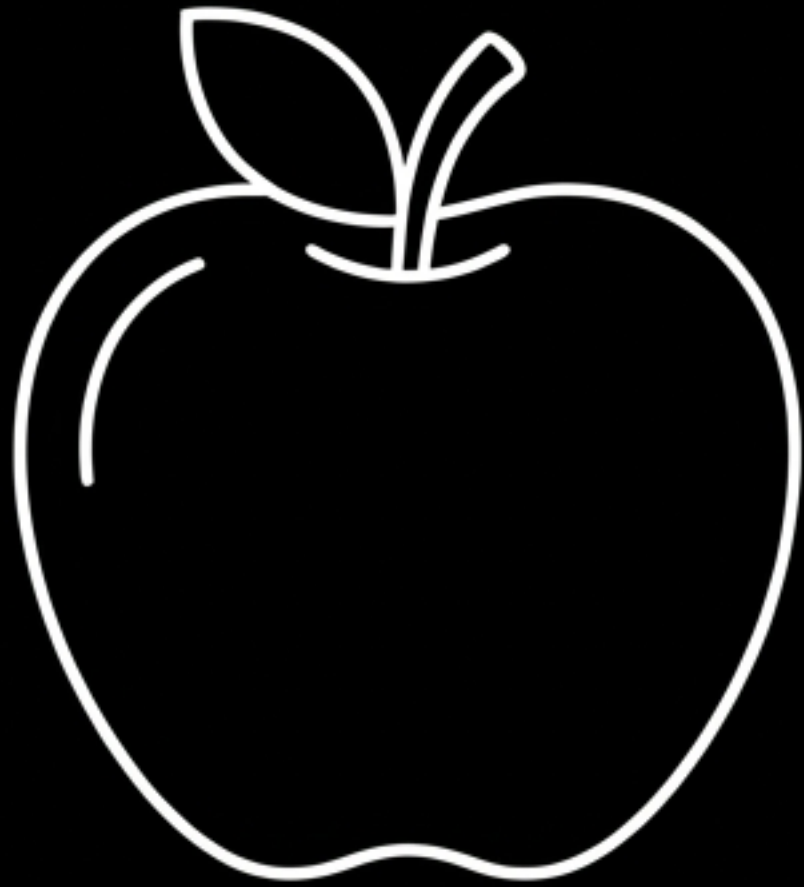


For the next 380,000 years, nothing much happened.
The universe was just an opaque, glowing fog of particles.

On Earth as in the Heavens



Until Isaac Newton, we believed Earth
and the Heavens obeyed different rules.
Newton breached this barrier.



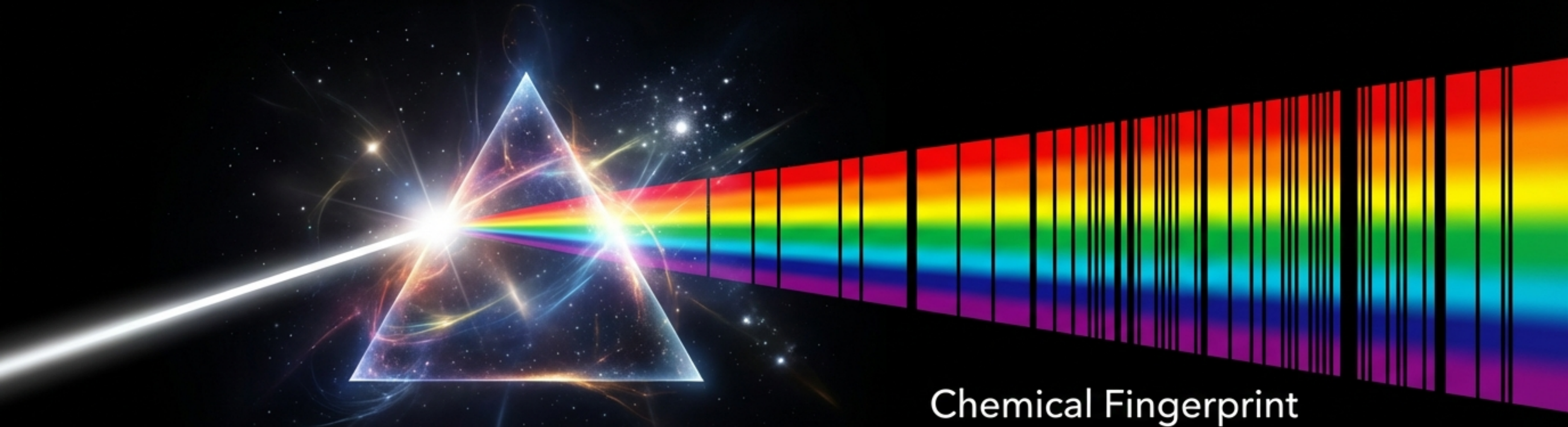
$$F = G \frac{m_1 m_2}{r^2}$$



The force that pulls a ripe apple from a tree is the same force that keeps the Moon in orbit.
This universality of physical laws drives scientific discovery like nothing else.

The Message in the Light

When we pass light through a prism, it reveals a spectrum. In the 19th century, a spectrum from the Sun revealed an element unknown on Earth.



Astronomers named it Helium (after Helios, the Sun). It was the first element discovered in space before it was found on Earth.

The Speed of Light

$$c = 299,792,458 \text{ m/s}$$

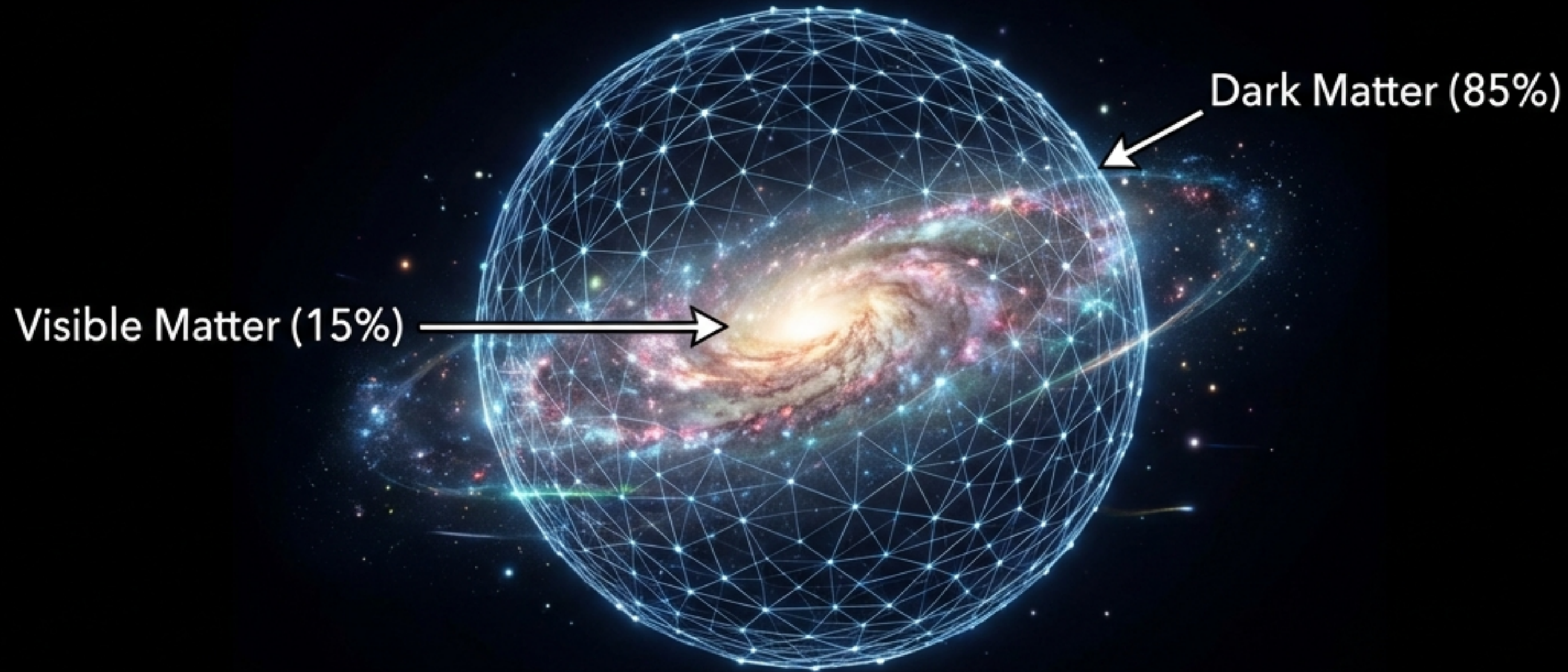
It's Not Just a Good Idea. It's the Law.

No experiment has ever revealed an object outrunning a beam of light. This isn't a challenge for engineers; it is a fundamental principle of causality.

Because light takes time to travel, looking out into space is literally looking back in time.

Conservation & Mystery

Conservation laws apply everywhere. However, we measure more gravity in the universe than visible matter can explain.

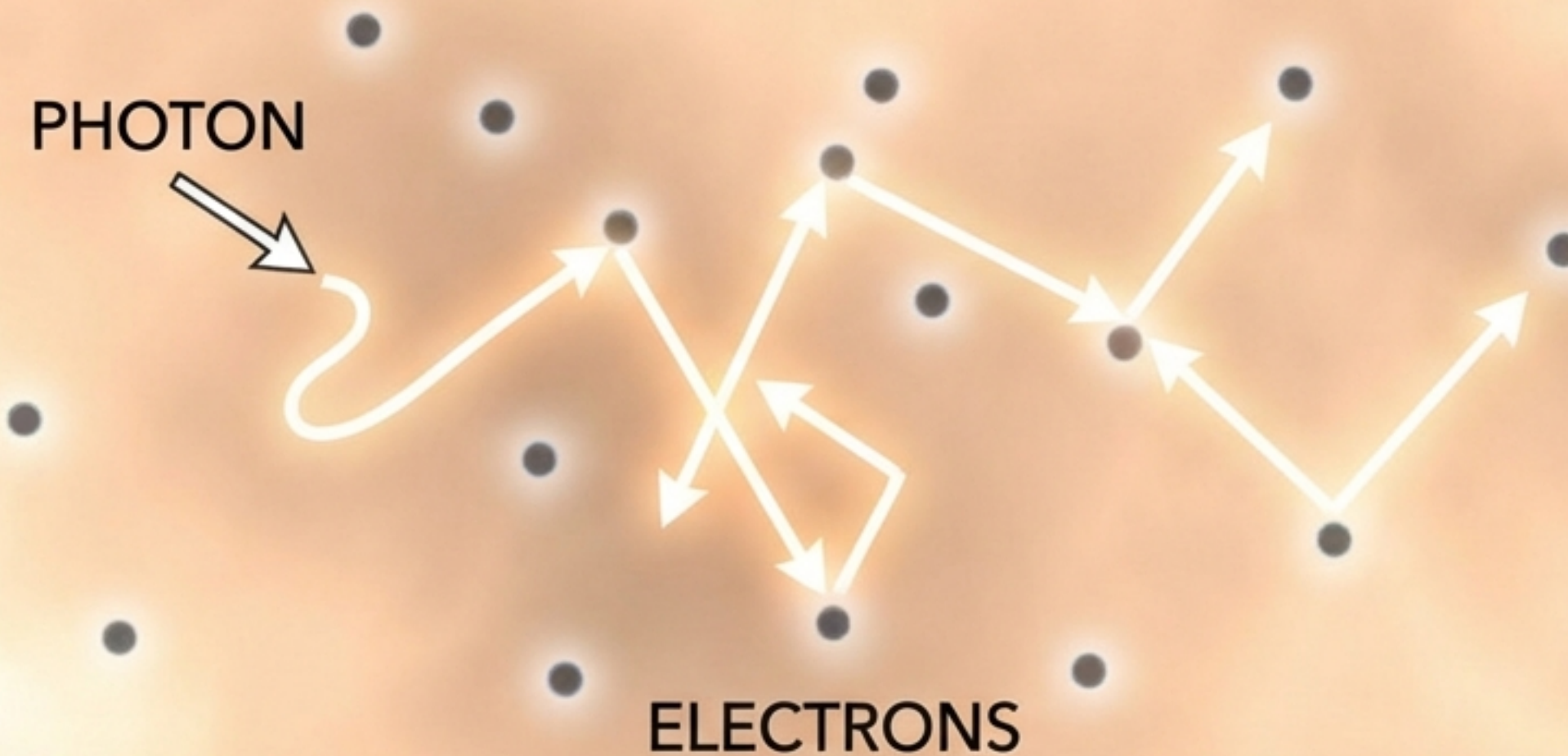


We call it Dark Matter. It has gravity, but it doesn't interact with light. We can't see, taste, or touch it, but we know it's there because the galaxy obeys the laws of gravity.

TIME: $t < 380,000$ years

The Fog

For millennia, the universe was an opaque soup. Photons (light) couldn't travel more than a nano-distance before hitting a free electron.



If you had been there, you couldn't see your hand in front of your face. Light was trapped.

TIME: $t = 380,000$ years

The Release

Recombination (3,000 Kelvin)

The temperature dropped enough for electrons to be captured by atomic nuclei (The Marriage). Photons were suddenly free to travel uninterrupted.

This ancient light is still traveling today.

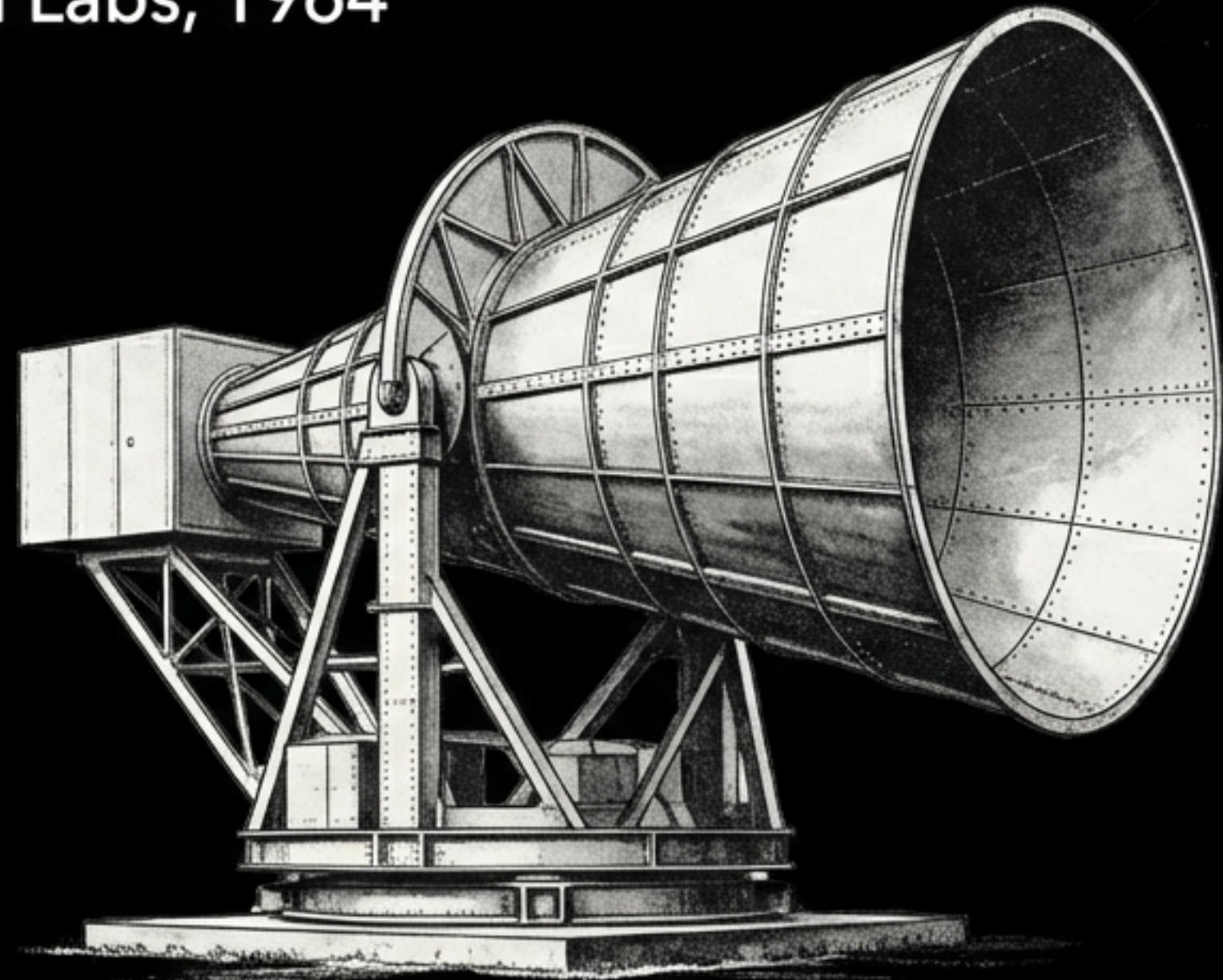
TIME: $t \approx 13.8$ billion years

THE PIGEON POOP INCIDENT

Bell Labs, 1964

Arno Penzias & Robert Wilson were trying to clean microwave communication signals. They found a persistent background hiss that wouldn't go away.

They scrubbed "white dielectric substance" (pigeon droppings) from the dish, but the signal remained.



They had accidentally discovered the Cosmic Microwave Background (CMB)—the afterglow of the Big Bang.

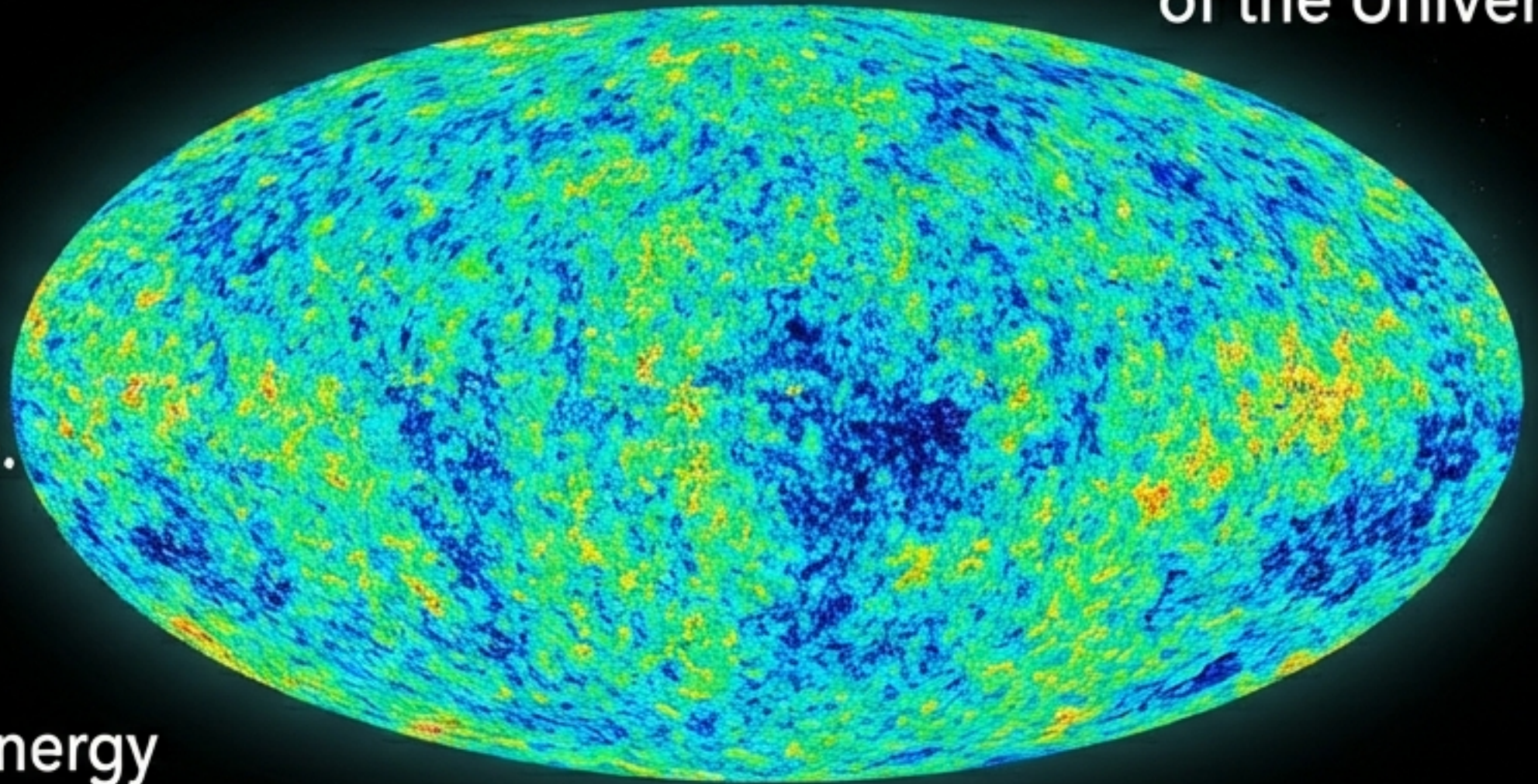
TIME: $t \approx 13.8$ billion years

COSMIC PHRENOLOGY

The Baby Picture
of the Universe

The CMB is a precise map of where matter was 380,000 years after the Big Bang. By analyzing the "bumps" in this map, we can deduce the structure of the infant universe.

- Age of the universe
- Strength of gravity
- Amount of dark matter and energy



TIME: $t \approx 13.8$ billion years

CHECKING THE TEMPERATURE

The CMB is 2.7 Kelvin today. But in the past, the universe was smaller and hotter. We can observe the molecule Cyanogen in distant galaxies billions of light-years away.

Hot

Past

Expansion & Cooling

2.7 Kelvin

Present

The Cyanogen in distant galaxies is excited by radiation warmer than ours. We are literally watching the universe cool down over time.

TIME: $t \approx 13.8$ billion years

THE GALAXY AND THE SEVEN DWARFS



THE EXCEPTION

When we look at the universe, we focus on beautiful spirals. But dwarf galaxies outnumber large galaxies by ten to one. They are small, dim, and 'sorry-looking,' yet they are the standard galactic form.

THE RULE (DWARF GALAXIES)

TIME: $t \approx 13.8$ billion years

GALACTIC CANNIBALISM



Dwarf galaxies often orbit larger ones, but it's a hazardous life. The Milky Way is currently consuming a dwarf galaxy whose remains are visible as a stream of stars.

"The system is called the Sagittarius Dwarf, but should probably have been named Lunch." — Neil deGrasse Tyson

TIME: $t \approx 13.8$ billion years

JUST HOW VOID IS THE VOID?

The space between galaxies is not empty. It is filled with:



HOMELESS STARS
Vagabond stars thrown
out of galaxies.

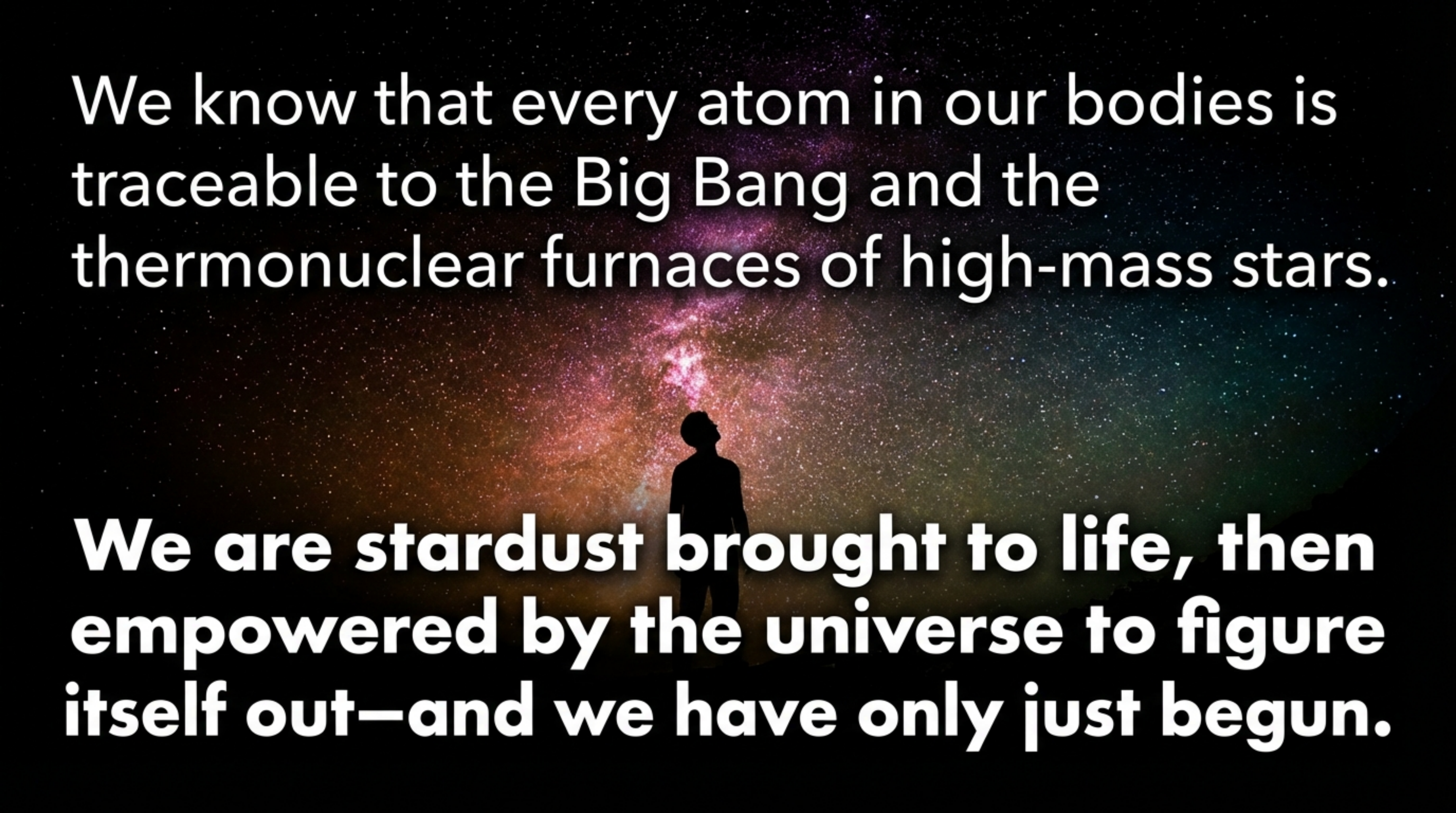


SUPERNOVAS
Exploding stars in the
middle of nowhere.

X-RAY GAS
Gas at millions
of degrees.



DARK MATTER
The bulk of the
universe's mass.

A silhouette of a person stands in the center, looking up at a vibrant, colorful nebula in space. The nebula features a bright, glowing core with shades of pink, purple, and orange, surrounded by a vast field of stars and cosmic dust. The background is a deep black, emphasizing the brilliance of the celestial phenomenon.

We know that every atom in our bodies is traceable to the Big Bang and the thermonuclear furnaces of high-mass stars.

We are stardust brought to life, then empowered by the universe to figure itself out—and we have only just begun.